

Ismail Ahmed

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EDUCATION

University of Toronto

Honours Bachelor of Science in **Computer Science**

Toronto, ON

- **Leadership:** Director at **UofT AI**; chaired technical panels with industry experts and managed a community of 200+ student developers to bridge the gap between academia and industry AI trends.
- **Relevant Coursework:** Data Structures & Algorithms, Software Architecture, Distributed Systems, Operating Systems, Database Management, Computer Networks, Cloud Computing.

EXPERIENCE

Automated DesignWorks

Jan. 2026 – Present

Software Developer

Toronto, ON

- Architected a modular **Object-Oriented automation framework** utilizing the **SolidWorks API**, implementing complex **computational geometry algorithms** to automate 3D modeling.
- Developed custom **JSON serialization algorithms** and **Stack-based data structures** to manage state configuration, enabling seamless bidirectional data interchange.
- Engineered **event-driven UI systems** for parametric design inputs, optimizing memory management and workflow logic to resolve runtime bottlenecks and enhance application stability.

University of Toronto

Jan. 2025 – May. 2025

Machine Learning Researcher

Toronto, ON

- Spearheaded research into **Generative Modeling**; trained **WGANs** and **DDM-CGANs** to synthesize financial time-series data, achieving a **98% distribution match** against historical tail-risk data.
- Quantified model reliability using **Monte Carlo simulations** and **Frobenius norms** for stress-testing.

LangChain-AI (Open Source Contributor)

Sep. 2024 – Dec. 2024

Software Developer

Remote

- Authored and upstreamed a **Dropbox document loader** module for the **LangChain.js** ecosystem, providing seamless cloud storage integration for the global developer community (30k+ stars on GitHub).
- Implemented robust error-handling and recursive file-parsing logic in **TypeScript**, ensuring stable data streaming for Large Language Model (LLM) ingestion pipelines.

Lifomation

Jan. 2023 – Dec. 2023

Software Developer

Toronto, ON

- Architected an enterprise **Retrieval-Augmented Generation (RAG)** system using **FAISS** and **OpenAI Embeddings**, enabling semantic search across 10,000+ docs with 90%+ retrieval precision.
- Engineered high-throughput ingestion pipelines in **Python**, optimizing vector database indexing which reduced query latency by **45%** and improved system response times for 500+ daily active users.
- Designed **MLOps monitoring** dashboards to track model drift and retrieval accuracy in real-time.

SELECTED PROJECTS

DevSync Real-time Collaborative Workspace | React, Node.js, Socket.io, Redis

- Developed a full-stack collaborative platform with **real-time synchronization** using **WebSockets**, allowing multiple users to edit code and documents simultaneously with **< 50ms** latency.
- Implemented **Redis-based caching** for session management and **JWT authentication** for secure access.
- Built a responsive UI with **Tailwind CSS** and integrated a custom Markdown compiler for live previews.

VisionEdge | Python, TensorFlow, OpenCV, Flask, Docker

- Developed a **lightweight CNN** optimized for edge devices, achieving real-time inference (30+ FPS) and a **92% mAP**. Containerized with **Docker** for seamless cross-platform deployment.

TECHNICAL SKILLS

Languages: Python (Expert), Java, C++, C#, JavaScript/TypeScript, SQL, Go, Bash, HTML/CSS, VBA

Frameworks: React, Next.js, Spring Boot, Node.js, Express, .NET Core, FastAPI, Flask, Django, GraphQL

Cloud/DevOps: AWS (EC2, Lambda, S3, RDS), Docker, Kubernetes, Terraform, Jenkins, GitHub Actions, Linux, Nginx

Databases: PostgreSQL, MongoDB, Redis, MySQL, DynamoDB, Elasticsearch, Cassandra, SQLite, Supabase

AI/ML: PyTorch, TensorFlow, Scikit-learn, LangChain, Hugging Face, OpenCV, RAG, Pandas, NumPy

Tools & Testing: Git, JIRA, Postman, Selenium, PyTest, JUnit, Jest, Agile/Scrum, System Design, Microservices